Properly Colored Trails, Paths, and Bridges

Wayne Goddard[1] and Robert Melville

Dept of Mathematical Sciences
Clemson University

Abstract

The proper-trail connection number of a graph is the minimum number of colors needed to color the edges such that every pair of vertices are joined by a trail without two consecutive edges of the same color; the proper-path connection number is defined similarly. In this paper we consider these in both bridgeless graphs and graphs in general. The main result is that both parameters are tied to the maximum number of bridges incident with a vertex. In particular, we provide for $k \geq 4$ a simple characterization of graphs with proper-trail connection number k, and show that the proper-path connection number can be approximated in polynomial-time within an additive 2.

Key-words: edge coloring, properly colored trails, path connection number, bridges

1 Introduction

We consider the problem of coloring the edges of a graph so that it is possible to get between any pair of vertices without two consecutive edges having the same color. Obviously, this can be achieved by giving every edge in the graph a different color and indeed by any proper coloring of the edges. So the real question is: What is the minimum number of colors one needs?

There are three associated parameters for a connected graph G. We define the *proper-walk connection number* $pW(G)$ as the minimum number of colors if one is allowed any properly colored walk. We define the *proper-trail connection number* $pT(G)$, where one is restricted to trails (no repeated edge). And we define the *proper-path connection number* $pP(G)$ where one is restricted to paths (no repeated vertex). Trivially, $pW(G) \leq pT(G) \leq pP(G)$. We will call a coloring *valid* if has the desired walks, trails, or paths.

[1]Corresponding author: goddard@clemson.edu

Now, the proper-path connection number has been studied by several authors under the name the *proper connection number*, as introduced by Borozan et al. (2012) and Andrews et al. (2016). For example, Borozan et al. (2012) showed that $pP(G) \leq 3$ for any bridgeless graph and $pP(G) \leq 2$ for any 3-connected graph. For a survey, see Li and Magnant (2015). We (Melville and Goddard, 2016) introduced the proper-walk connection number. In particular, we showed that $pW(G) \leq 3$ for any connected graph G that is not a tree, and that $pW(G) \leq 2$ if G is bridgeless. The proper-trail connection number has not been studied.

In this paper we investigate the three parameters with emphasis on the proper-trail connection number. The main result is that both the proper-trail and proper-path connection numbers are tied to the maximum number of bridges incident with a vertex. In particular, we provide for $k \geq 4$ a simple characterization of graphs G such that $pT(G) = k$. Also, we show that $pP(G)$ can be approximated in polynomial-time within an additive 2.

We proceed as follows. In Section 2 we provide examples to compare and contrast the three parameters. We investigate the proper-trail connection number for bridgeless graphs in Section 3 and for general graphs in Section 4. Some thoughts on the analogous question for the proper-path connection number are given in Section 5. We conclude with a brief comment on the directed version in Section 6 and ideas for further work in Section 7.

2 Examples and Comparisons

Recall that $pW(G) \leq pT(G) \leq pP(G)$ for all graphs G. Trivially the three parameters are all 1 if G is complete, and all at least 2 otherwise.

We show first that both inequalities are proper. Indeed, the gap between $pT(G)$ and $pP(G)$ can be arbitrarily large. For example, take a star on m edges and add one edge; call the the resultant graph G_m. For example, G_6 is shown in Figure 1. Then $pT(G_m) = 2$ (color red the two edges in the triangle incident with the center and color blue the remaining edges), while $pP(G_m) = m - 1$.

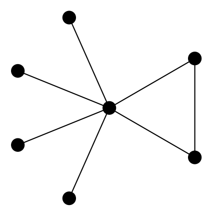

Figure 1. *A graph G_m where $pT(G_m) \ll pP(G_m)$*

Also, the gap between $pW(G)$ and $pT(G)$ can be arbitrarily large. For example, take a star on m edges and a triangle, and identify an end-vertex of the star and a vertex of the triangle; call the resultant graph H_m. For example, H_5 is shown in Figure 2. Then $pW(H_m) = 2$ (color blue the central edge and the edge of the triangle not incident with the start and color red the remaining edges), while $pT(H_m) = m$.

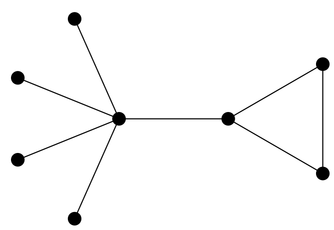

Figure 2. *A graph H_m where $pT(H_m) \gg pW(H_m)$*

On the other hand, there are cases of equality. For example, the following result follows since a trail in a graph of maximum degree at most 3 must be a path:

Lemma 1 *If graph G has maximum degree at most 3, then $pP(G) = pT(G)$.*

We (Melville and Goddard, 2016) characterized the bipartite graphs G that have $pW(G) = 2$ and observed that the same characterization holds for $pP(G) = 2$. By the same reasoning (or because $pT(G)$ is sandwiched between $pW(G)$ and $pP(G)$), the same characterization holds for the proper-trail connection number. To state the result, define $M(G)$ as the spanning subgraph that results if one removes all the bridges of graph G. (Note that each component of $M(G)$ is either an isolated vertex or is 2-edge-connected.) From (Melville and Goddard, 2016) we thus obtain:

Theorem 1 *Let G be a noncomplete connected bipartite graph with order at least 3. Then $pT(G) = 2$ if and only if every component of $M(G)$ is incident with at most two bridges.*

Moreover, if G is bridgeless, then one can find a 2-coloring such that every pair of vertices are joined by a properly colored path starting with any designated color.

3 Trails in Bridgeless Graphs

Borozan et al. (2012) proved that:

Theorem 2 *If G is a connected bridgeless graph, then $pP(G) \leq 3$.*

It is easy to find bridgeless graphs that require three colors for properly colored paths. For example, consider the bowtie graph B_m formed by taking m copies of the triangle and identifying one vertex on each. (Equivalently, B_m is the join of K_1 and mK_2.) Then:

Lemma 2 *(a) $pT(B_m) = 2$ for $m \geq 2$.*
(b) $pP(B_2) = 2$ and $pP(B_m) = 3$ for $m \geq 3$.

Proof. (a) This follows since the bowtie is a supergraph of the graph in Figure 1. Or note that it has an Eulerian tour.

(b) For B_2, the same coloring as the trail case from (a) works. For the lower bound it suffices to verify it for B_3. This can be done by hand or by computer, and is also given in (Huang et al., 2016). The upper bound follows from Theorem 2; but an explicit coloring is to properly color each triangle. □

It is harder to find examples of 2-connected graphs where the proper-path connection number is 3. Borozan et al. (2012) gave the example shown in Figure 3. By Lemma 1, this graph has proper-trail connection number 3 also.

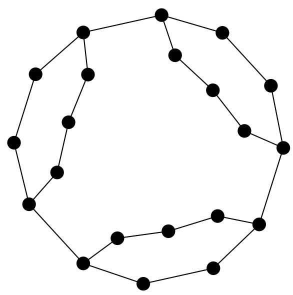

Figure 3. *A block B where $pT(B) = 3$*

We next consider the general idea of appending cycles to an existing graph.

Lemma 3 *Let G be a connected graph with vertex v in some cycle. Let H be the graph that results from adding a disjoint bipartite block B and identifying one vertex of B with v. Then if $pT(G) \in \{2, 3\}$ it follows that $pT(H) = pT(G)$.*

PROOF. We show first that $pT(H) \leq \max\{pT(G), 2\}$. As per Theorem 1, one can color the block B with two colors so that every pair of vertices are joined by a properly colored path starting with each color. This extends the coloring of G to H.

Conversely, we show that $pT(H) \geq \min\{pT(G), 3\}$. Suppose the proper-trail connection number of H is two. Consider a properly colored trail T between vertices x and y of G. Every closed trail in B has even length. So if the trail T uses B, reaching v the first time on say a red edge, then it will reach v again on a red edge after using B. That is, B is of no use: it must be that the coloring restricted to G is a valid coloring that shows that $pT(G) \leq 2$. $\square$

Actually the limitation on $pT(G)$ is unnecessary; see the discussion after Corollary 1.

Adding nonbipartite blocks, on the other hand, is a much different story. For example, define the *generalized corona* $cor(G, F)$, for F a rooted graph, as follows. Take the graph G and, for each vertex $v \in G$, introduce a copy of F and identify the root of that copy with v. An example is shown in Figure 4.

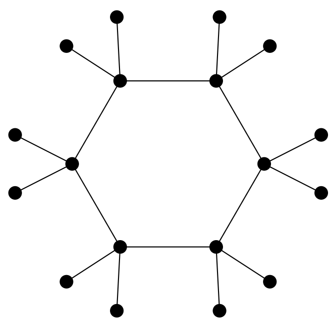

Figure 4. *The graph $cor(C_6, P_3)$ (for P_3 rooted at its center)*

Lemma 4 *Let G be a nontrivial connected graph. Then $pT(cor(G, K_3)) = 2$.*

PROOF. The resultant corona is noncomplete; so we need at least two colors. For a valid coloring, color red every edge of G, as well as the edge of each K_3 not incident with its root; then color blue the two edges of each K_3 incident with the root. Every x–y path in the uncolored G corresponds to a properly colored x–y trail in the colored corona by, at every vertex of G, traversing the K_3. $\square$

This idea provides examples of bridgeless graphs where trails and paths require different numbers of colors. Specifically, take a bridgeless graph G such that $pP(G) = 3$ (for example, the graph in Figure 3 above). Then $pP(cor(G, K_3)) = 3$, but $pT(cor(G, K_3)) = 2$. Also, the same idea works if one replaces the K_3's with any collection of odd cycles.

We are unable to find a block where the two parameters differ. All examples needing 3 colors for the proper-path connection number given by Huang et al. (2017a) and Brause et al. (2017) also need 3 colors for trails. One avenue we have considered is regular graphs. If a connected graph is r-regular for r even then it is Eulerian. And if a graph has an Eulerian trail, then the proper-trail number is at most two: simply color the trail with alternating colors. So that raises the question about whether there are r-regular blocks for r even that require three colors for properly colored paths. But we do not know the answer to that.

The following result, stated for proper-path connection number two, is given by Borozan et al. (2012). The result immediately holds for trails and walks, and without restriction on the number of the original graph. The proof is the same as in the original paper.

Lemma 5 *Let G be a noncomplete connected graph and let graph G' be formed by adding a vertex x and joining x to at least two vertices of G. Then $pW(G') \le pW(G)$ and $pT(G') \le pT(G)$ and $pP(G') \le pP(G)$.*

Proof. Say vertex x is added with neighbors u and v. Since G is noncomplete, it requires at least two colors. In this coloring of G there is a properly colored connection (meaning walk, trail, or path as appropriate) P between u and v. Color the edge xu different to the first edge of P and the edge vx different to the last edge of P. This provides a suitable connection between x and any vertex on P. Consider getting from x to some vertex w not on P. Then there is a properly colored connection from w to (say) u. Let z be the first vertex on P that is encountered. Then there are two connections from x to z, one via u and one via v, and they end in different colors. So one of these connections can be combined with the z–w connection to get to w with a properly colored connection. $\square$

4 Trails in Graphs with Many Bridges

Several authors (e.g. Huang et al. (2015)) have observed that the proper-path connection number in a graph with bridges can be reduced to focussing on the bridgeless pieces. The same is true for trails. We formalize this as follows. As before, let $M(G)$ be the spanning subgraph of G that remains after the bridges of G have been removed. Then for any component X of $M(G)$, define $\mathit{Fuzz}(X)$ to be the subgraph of G induced by the closed neighborhood $N[V(X)]$; that is, it is the subgraph consisting of X and all bridges incident with X.

Lemma 6 *For any connected graph G, it holds that $pT(G) = \max\{\, pT(\mathit{Fuzz}(X)) : X \text{ a component of } M(G)\,\}$. A similar result holds for paths.*

Proof. Any bridge can be used at most once in any trail. Thus the coloring of G restricted to each $\mathit{Fuzz}(X)$ is a valid coloring thereof. Conversely, given valid colorings of each $\mathit{Fuzz}(X)$, one can combine them by permuting the colors if necessary. $\square$

We will need the following parameter. For a vertex v, define $b(v)$ as the number of bridges incident with v. Then define:

> $B_{iso}(G)$ is the maximum of $b(v)$ over all vertices v that do *not* lie in cycles.

That is, out of all vertices that are incident only with bridges, this quantity is the maximum degree thereof. (We define $B_{iso}(G) = 0$ if all vertices in G lie in cycles.) It is immediate that $pT(G) \geq B_{iso}(G)$, since if v does not lie in a cycle its incident edges must receive different colors.

We show that if $B_{iso}(G)$ is sufficiently large, then it determines the proper-trail connection number. We will need the following extension of Theorem 2. Indeed, we claim that the proof by Borozan et al. (2012) also establishes this extension.

Lemma 7 *If graph G is 2-edge-connected, then the edges of G can be colored with 3 colors, and each vertex w assigned a free color f_w, such that:*

(1) For all vertices u_1 and u_2, there are two properly colored u_1–u_2 paths such that (a) their start colors are different; (b) their end colors are different; and (c) for at least one $i \in \{1, 2\}$ neither edge at u_i uses color f_{u_i};

(2) For all vertices u, there is a cycle containing u that is properly colored and neither edge at u has color f_u.

Proof. Consider the proof of Theorem 2 given in (Borozan et al., 2012) (where it is Theorem 4). We claim that their inductive proof also shows this result. Their proof is by ear decomposition. The existence of a properly colored cycle containing each vertex is explicitly mentioned in the proof, as are parts (a) and (b) of Condition 1 (sometimes called the "strong property"). When an ear is added, each new vertex w has only two incident edges. So one can define f_w to be the color missing at w when it is added. Condition 1(c) follows as does the full Condition 2. $\square$

Note that in particular, Condition 1 implies that for all vertices u_1 and u_2, there is a properly colored u_1–u_2 path that starts with a color not f_{u_1} and ends with a color not f_{u_2}. As a consequence of Lemma 7 we obtain:

Theorem 3 *If G is a connected graph, then $pT(G) \leq \max\{B_{iso}(G), 3\}$.*

Proof. By Lemma 6, it suffices to show that $pT(cor(H, F)) \leq 3$, where H is any 2-edge-connected graph and F is any star rooted at its center. So, take H and color it with 3 colors by Lemma 7. Then, for every vertex w of H, color every leaf incident to w with the free color f_w.

We claim that this coloring is valid. One can enter and leave H via bridges at the same vertex w by using the cycle inside H guaranteed by Condition 2. This handles two end-vertices of the corona that do have a common neighbor. Further, two end-vertices of the corona that don't have a common neighbor are joined by a properly colored path by Condition 1. $\square$

It follows that if $B_{iso}(G) \geq 3$, then $pT(G) = B_{iso}(G)$. Indeed we have the following characterization:

Corollary 1 *For $k \geq 4$, the proper-trail connection number of a connected graph G is k if and only if $B_{iso}(G) = k$.*

Note that this characterization also shows that Lemma 3 applies without the limitation on $pT(G)$.

5 Paths in Graphs with Many Bridges

For the path parameter, one can also readily provide a narrow range for the parameter.

We will need the following definitions. Recall that $b(v)$ is the number of bridges incident with vertex v, and $B_{iso}(G)$ is the maximum of $b(v)$ over all vertices v that do not lie in cycles. Now, define:

$B_{cyc}(G)$ is the maximum of $b(v)$ over all vertices v that do lie in cycles.

Let $B(G) = \max(B_{iso}(G), B_{cyc}(G))$. As Andrews et al. (2016) observed, it holds that $pP(G) \geq B(G)$. Note that $B(G)$ is not a lower bound for the proper-trail connection number (as for example, the graph in Figure 1 shows).

It is immediate from Lemma 6 and Theorem 2 that $pP(G)$ is at most the quantity $\max\{B_{iso}(G), B_{cyc}(G) + 3\} \leq B(G) + 3$, by coloring the bridges properly using a palette disjoint from the colors for (the nontrivial components of) $M(G)$. But this bound can be improved slightly:

Theorem 4 *For all connected graphs G, it holds that $pP(G) \leq \max\{B_{iso}(G), B_{cyc}(G)+2, 3\}$.*

Proof. By Lemma 6, it suffices to show that $pP(cor(H, S_M)) \leq m + 2$ where H is any 2-edge-connected graph and S_m is the star with m edges rooted at its center. So, take H and color it with a trio $\mathcal{T}$ of 3 colors by Lemma 7. Then, for each vertex w of H, color the m leaves incident to w each with a different color using neither color in $\mathcal{T} - \{f_w\}$. By Condition 1 this coloring is valid. □

The above result is a slight improvement on Theorem 6 of (Borozan et al., 2012): namely that $pP(G) \leq \max\{\widetilde{\Delta}, 3\}$, where $\widetilde{\Delta}$ is the maximum degree of a vertex that is incident with at least one bridge. Note that $\widetilde{\Delta} \geq \max\{B_{iso}(G), B_{cyc}(G) + 2\}$.

It also follows that:

Corollary 2 *For all connected graphs G with bridges, it holds that $B(G) \leq pP(G) \leq B(G) + 2$.*

Corollary 2 is best possible in that there are graphs with $B(G)$ arbitrarily large and each of $pP(G) = B(G)$, $pP(G) = B(G) + 1$, and $pP(G) = B(G) + 2$.

An example for the first case is a star. More generally, we have the following observation:

Lemma 8 *If graph H is 3-edge-connected, then for all B sufficiently large it holds that $pP(cor(H, S_B)) = B$ where S_B is the star on B edges rooted at its center.*

PROOF. Let $B \geq \chi'(H)$. Then color the edges of H properly. Further, for each vertex w of H give the bridges at w different colors. Then it is possible to get between any two end-vertices x and y of the corona whose neighbors are distinct, since between their neighbors there are three paths, each starting and finishing with a different color. $\square$

For the case that $pP(G) = B(G) + 1$, take the graph drawn in Figure 1. For the case that $pP(G) = B(G) + 2$, consider the generalized corona $cor(C_n, S_B)$. An example was given in Figure 4. Then the following is a special case of Corollary 4.5.3 of Laforge (2016):

Theorem 5 *For all $B \geq 1$ and $n \geq 4$, it holds that $pP(cor(C_n, S_B)) = B + 2$ where S_B is the star on B edges rooted at its center.*

Now, Huang et al. (2015, 2017b) recently provided another upper bound. Specifically they define for graph G, the value $\Delta(G^*)$ as the maximum number of bridges incident with any element of $M(G)$. They then show that $pT(G) \leq \max\{\Delta(G^*), 3\}$. This bound can be both better and worse than Theorem 4. So we present next a mutual improvement.

Consider each nontrivial component H_i of $M(G)$ and sort the vertices $v_1, v_2, \ldots$ in nonincreasing order by their b-value. Then define $B'(H_i) = \max\{b(v_1), b(v_2) + 2\}$.

Theorem 6 *For all connected graphs G it holds that*
$pP(G) \leq \max\{B'(H_1), \ldots, B'(H_k), B_{iso}(G), 3\}$.

PROOF. The proof is similar to the proof of Theorem 4. The only change is that when one colors the bridges incident with v_1, one is allowed to use all colors. By Lemma 7, one can extend a path entering at v_1 to any other vertex w of H (the strong property). And then one can leave w on a bridge of any color except $\mathcal{T} - f_w$, as before. $\square$

This result is close to best possible in that Laforge (2016) provided an extension to Theorem 5; namely that if one takes the cycle C_n for $n \geq 4$ and adds B leaves incident with each of any three vertices, then the resultant graph requires $B + 2$ colors.

6 Directed Graphs

For a strongly connected digraph D, one can define the proper-walk, proper-trail, and proper-path connection numbers as in the undirected case. Magnant et al. (2016) introduced the idea for paths, and we (Melville and Goddard, 2016) introduced the idea for walks. In particular $pW(D) \leq pP(D) \leq 3$, and there are graphs where $pW(D) = 2$ and $pP(D) = 3$.

We observe that for a properly-colored walk in a directed graph, there is no reason to use an arc more than once. That is, two vertices are joined by a properly-colored walk if and only if they are joined by a properly-colored trail. Thus: For all strongly-connected digraphs D, it holds that $pW(D) = pT(D)$.

7 Future Work

We conclude with some thoughts on future work. The most glaring open problem is the complexity of recognizing which graphs have the parameter 2. Is there a polynomial-time algorithm, or is it NP-hard? Note that Ducoffe et al. (2017) recently showed that the proper-path connection number is NP-hard for directed graphs.

One specific family of graphs that seems worthy of study is regular graphs. For even degree such graphs are Eulerian and so only two colors are needed for properly colored trails. But what about odd degree? And all degrees for properly colored paths? In a different direction, Andrews et al. (2016) showed that the Cartesian product of two nontrivial connected graphs has $pP(G) = 2$. (This also follows from the fact that the Cartesian product of two nontrivial trees is 2-connected and bipartite.) Recall that a *generalized prism* (or permutation graph) of graph G is obtained by taking two copies of G and adding a matching between the two copies. Andrews et al. (2016) ask whether there is a generalized prism H with $pP(H) \geq 3$.

Another direction of interest is the question where some of the edges of the graph are already colored. For example, Kézdy and Wang (1999) asked when one could complete a 2-coloring such that there is a properly colored path between two specified vertices.

References

Andrews E, Lumduanhom C, Laforge E, and Zhang P (2016) On proper-path colorings in graphs. J Combin Math Combin Comput 97:189–207

Borozan V, Fujita S, Gerek A, Magnant C, Manoussakis Y, Montero L, and Tuza Z (2012) Proper connection of graphs. Discrete Math 312:2550–2560

Brause C, Doan T, and Schiermeyer I (2017) Minimum degree conditions for the proper connection number of graphs. Graphs Combin 33:833–843

Ducoffe G, Marinescu-Ghemeci R, and Popa A (2017) On the (di)graphs with (directed) proper connection number two. See hal-01491146

Huang F, Li X, Qin Z, and Magnant C (2016) Minimum degree condition for proper connection number 2. To appear in Theoret. Comput. Sci.

Huang F, Li X, Qin Z, Magnant C, and Ozeki K (2017a) On two conjectures about the proper connection number of graphs. Discrete Math 340:2217–2222

Huang F, Li X, and Wang S (2015) Proper connection number and 2-proper connection number of a graph. ArXiv:1507.01426

Huang F, Li X, and Wang S (2017b) Upper bounds of proper connection number of graphs. J Comb Optim 34:165–173

Kézdy A and Wang C (1999) Alternating walks in partially 2-edge-colored graphs and optimal strength of graph labeling. Discrete Math 194:261–265

Laforge E (2016) Chromatic Connectivity of Graphs. Thesis (Ph.D.), Western Michigan University

Li X and Magnant C (2015) Properly colored notions of connectivity—a dynamic survey. Theory and Applications of Graphs 0:Article 2

Magnant C, Morley P, Porter S, Salehi Nowbandegani P, and Wang H (2016) Directed proper connection of graphs. Mat Vesnik 68:58–65

Melville R and Goddard W (2016) Coloring graphs to produce properly colored walks. To appear in Graphs Combin. See arXiv:1610.04874